

SAIDEEP KATRAGADDA

512-648-7227 | sdkatragadda@gmail.com | [linkedin.com/in/saideep-katragadda](https://www.linkedin.com/in/saideep-katragadda) | github.com/sydip | [sydip/Portfolio](#)

Education

Texas A&M University

College Station, TX

Bachelor of Science in Computer Science

Expected May 2028

- **Organizations:** Aggie Data Science Club (ADSC), Maroon Investment Club
- **Relevant Coursework:** Python Computation, Business Information Systems Concepts, Discrete Structures, Program Design and Concepts, Data Structures and Algorithms, Principles of Statistics, Programming Languages

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, FastAPI, Tailwind CSS, PyTorch/TensorFlow, Scikit-Learn, Pandas, Seaborn

Tools & Infrastructure: Docker, PostgreSQL, Git/GitHub, Postman, Azure DevOps, VS Code, Eclipse

Certifications: IT Specialist - Java (Certiport), Microsoft Office - Word, Excel, PowerPoint, Access

Experience

Software Engineer Team Lead

October 2026 – Present

American Airlines × ADSC

College Station, Texas

- Leading a **6-person team** to build a **time-series forecasting model** that predict airline market demand up to **2 years ahead** for **American Airlines Network Planning**, using GDD and demographic data.
- Directing **feature engineering** and **regression analysis** to identify drivers of market demand, and overseeing backtesting against historical and test data at the market, state, and entity level.

Undergraduate Researcher

October 2026 – Present

Texas A&M University

College Station, Texas

- Conducting research on an empirically informed **agent-based model** in **Python** and **NetLogo** that simulates how construction workers and crews decide over time whether to adopt occupational exoskeletons.
- Parameterizing the model from published literature and expert input, and running **simulation experiments** and **sensitivity analysis** to identify which factors drive long-term adoption versus discontinuation.

Application Developer Intern

June 2026 – August 2026

University Federal Credit Union (UFCU)

Austin, Texas

- Built AI-powered automation agents in **Azure DevOps** for sprint management and backlog tracking, plus a meeting intelligence pipeline that transcribed Microsoft Teams meetings and published AI-generated summaries to project wikis.
- Diagnosed and fixed failed lending requests in a banking loan **imaging API**, and built workflow automation with **Azure DevOps APIs** and **Azure cloud services** to cut project management work and improve documentation accuracy.

Quantitative Developer

March 2026 – Present

Maroon Investment Club

College Station, Texas

- Engineered a **dual-model XGBoost** classification system in **Python** to detect buy and sell opportunities during market dislocations, delivering a **36% backtested return** versus a **14% S&P 500** benchmark.
- Built a **feature-engineering pipeline** using **RSI**, **VIX**, moving averages, and momentum indicators, and applied **SMOTE** class balancing with **walk-forward validation** to reduce overfitting and improve out-of-sample performance.

Projects

CourtVision | Python, JavaScript, TypeScript, PostgreSQL, FastAPI, Next.js, Docker

2026

- Constructed a full-stack NBA analytics platform ingesting **131,000+ player game logs** across **6,151 games** and **26 draft classes**, with raw-cached, validated, and resumable data pipelines.
- Built a **random-forest prediction engine (19 models)** and **10,000-run Monte Carlo Finals simulation**, ranking actual MVP/ROY top 3 in **80% of backtested seasons**, with **94+% player similarity accuracy** and **95% AI query-routing accuracy**.

Gridiron | Python, Pandas, Scikit-Learn, Seaborn

2026

- Engineered a full-stack NFL against-the-spread analytics platform ingesting **489,000+ plays** across **2,911 games** with validated, idempotent pipelines and a **737-test suite** guaranteeing byte-identical served data.
- Designed a **leakage-safe logistic-regression model (11 engineered features)** validated via **7-fold walk-forward testing** on **1,828 out-of-sample games**, and caught a join fault dropping **181 team-games** of EPA/pace data.

Pixel Pathology (ADSC) | Python, PyTorch/TensorFlow, CNNs

2026

- Designing an interpretable **CNN** medical image classifier in **PyTorch** using **transfer learning** and **data augmentation**, with **Grad-CAM heatmaps** highlighting the regions that drive each prediction.